Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

Analysis of financial markets using machine learning

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Marek Šugár

Comenius University in Bratislava
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Department:	Department of Applied Mathematics and Statistics
Supervisor:	Mgr. Samuel Rosa, PhD.

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Marek Šugár

17424822

Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

THESIS ASSIGNMENT

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Annotation: Modeling and prediction of stocks in financial markets have long been a researched problem. Historically, this problem was approached using specific methods of financial mathematics, but with the significant development and success of machine learning in general, machine learning methods have also begun to be applied to predicting the behavior of financial markets. The aim of this thesis is to create a prediction model for selected stocks, which involves the appropriate selection and identification of relevant explanatory variables, as well as the suitable choice of model. Subsequently, it will be possible to analyze whether the obtained predictions can be used for successful creation of an investment portfolio.

Supervisor:	Mgr. Samuel Rosa, PhD.
Department:	FMFI.KAMŠ - Department of Applied Mathematics and Statistics
Head of department:	doc. Mgr. Igor Melicherčík, PhD.
Assigned:	02.10.2025
Approved:	28.10.2025

doc. Mgr. Tomáš Vinař, PhD.
Guarantor of Study Programme

.....
Student

.....
Supervisor

Declaration: I hereby declare that I have independently written this entire bachelor thesis entitled "Analysis of financial markets using machine learning", including all its figures and appendices, using the literature listed in the attached bibliography.

In preparation of the thesis I have also used the LLM ChatGPT for the keyword-based searches in particular Python libraries' documentations. I have used artificial intelligence tools in accordance with relevant legal regulations, academic rights and freedoms, ethical and moral principles while simultaneously maintaining academic integrity. I am aware that I am fully responsible for the accuracy of the final text.

Acknowledgments: This way I would like to thank my supervisor Dr. Rosa, for his mentorship, spent time and intriguing feedback throughout the whole process of thesis' creation. Last but not least, I would like to express thanks to all my family for their endless support and motivation.

Abstrakt

Na tomto mieste sa bude nachádzať slovenský abstrakt

Kľúčové slová: strojové učenie, akciové trhy, prediktívna analýza

Abstract

Here will be English written abstract.

Keywords: machine learning, stock markets, predictive analysis

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Introduction

Here will be the Introduction chapter, where reader will be introduced to the topic of Quantitative Science, its novelty and to the topics being dealt with in academic community and practice as well.

Chapter 1

Machine learning foundations

In this chapter, we aim to introduce methods of machine learning, which are subsequently used for the purposes of predictive modelling of selected set of stocks from the US stock market – particularly *Information technology* segment.

1.1 Machine learning as a method

1.2 Linear regression

1.3 Regularized variants of linear regression

1.4 k-Nearest neighbour algorithm

1.5 Random forest Algorithm

1.6 Time-series neural network (LSTM)

Chapter 2

Stock market data

The aim of this chapter is introduce set of related variables to the financial market, their possible importance in terms of predictive modelling and lastly their subsequential processing into specific forms, which were used through all the related analyses in this thesis.

Market itself and particularly price movement of financial instruments is a complex and unsurprisingly counterintuive process with a large number of more or less related variables, which contribute to the price movement. In terms of using such data, as an auxiliary source, for decision making in investment strategy creation, there is a rich universe of sources dealing with the issue [4], [7].

In the thesis, we aim mainly to research predictive power of various more or less sophisticated methods of machine learning on the particular market segment. Many papers tend to shift focus on single stocks or very limited set of a few stocks [5], [14]. Other tend to apply predicting modelling methods on stock funds *(e.g. US S&P 500 and NASDAQ or German DAX)* [11] however, there is a little number of sources dealing with a market segment as a whole and focus on single stocks included in it.

Our main focus is on the index fund S&P 500 – US stock index accumulating 500 biggest publicly traded companies in terms of market capitalization. The fund itself is considered to be the most trustworthy general indicator of the overall US and world's stock economy state [6]. The fund itself is divided into several general market segments to which stocks belong. In this thesis we focused on segment **Information technology** however, approach used in our analyses is adaptable to any subset of stocks – the only determining constraint is the respective database of stocks.

Information technology segment consists of 68 US-traded stocks. As of late 2025, in terms of total capitalization, this group makes total of nearly 30 % of the fund, which makes it the most weighted sector, despite making only 13.6 % of all included stocks [9]. This recent growth of its weight could be explained by a lots of factors steering the market nowadays, but the general view on the topic suggests, that recent boom

of artificial intelligence could be the most dominant engine of growth. This suggests that focusing on this segment is wise as it could be considered the most important one nowadays.

2.1 Variables in financial data

At first, we introduce specific attributes which define value movement of all financial and economical instruments used throughout the thesis as feature and target variables. These attributes could be further generalised onto any financial instrument attribute's movement, not only on price itself – as it will be applied on non-stock data as well.

For the purposes of defining desired terminology, let's define time interval on which we analyse instrument's attribute movement as t_1 till t_2, where naturally $t_1 < t_2$. Defined atttributes, which are attached to single time breakpoints, could be universally defined for any time interval with various frequency.

At the breakpoint t_1 we define **opening value**, which denotes value of the attribute at the very beginning of the interval. For the purposes of this thesis, where we aim to use daily frequency of data, this value denotes value of attribute at the very start of the trading day. In the opposite manner, at the breakpoint t_2 we define **closing price**, denoting the final value of the attribute at the end of the observed interval – in our case at the end of the trading day.

During the process of value evolving throughout the interval we observe global minimum and maximum of the attribute's value, which we denote **low value** and **high value** respectively. Such data give us scent of volatility during the interval and subsequently sentiment of the market regarding the attribute.

Last but not least we accumulate data regarding number of nominal instrument's stocks traded *(either sold or bought)* during the time interval. This is denoted as **volume** and in terms of the thesis this data is > 0 only for the stocks' data, as for some other instruments this data is irrelevant and not tracked anyhow.

Last data attribute, applied only on stocks' data as well, which is strongly dependent on single stocks is volume of **dividends** paid off. Some companies oblige themselves to pay off parts of their profit to their investors on regular basis *(e.g. Apple, Microsoft or Oracle)* [2]. With respect to the completness of data fetched we included this data in out potential features database.

Visualisation of the terms defined on sample interval of Microsoft stock between the beginning of June 2025 and beginning of September 2025 is pictured on Figure 2.1.

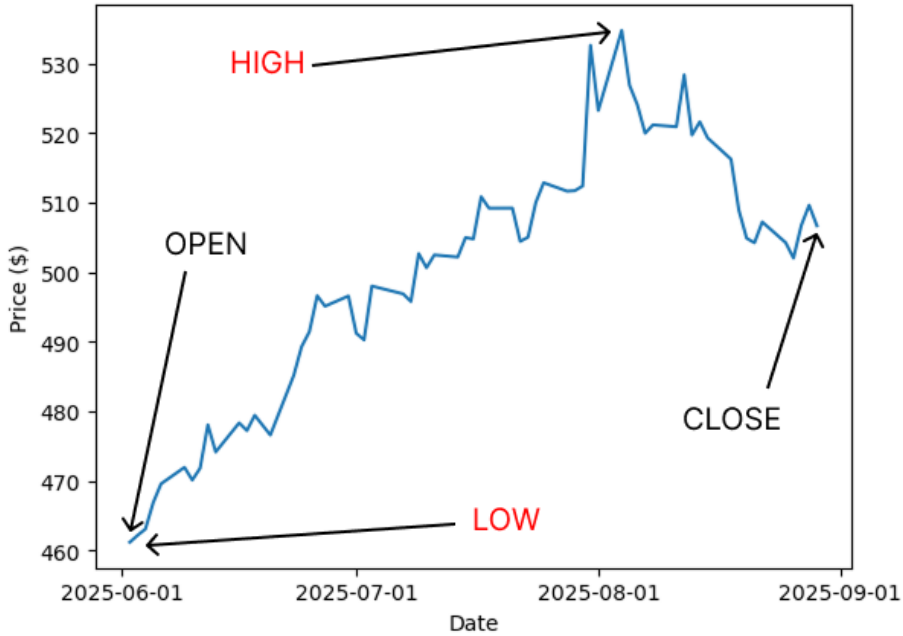

Figure 2.1: *Visualisation of the defined attribute's movement terms on Microsoft stock price.*

2.2 Rolling window methodology

There are quite a few approaches on how to properly process time-series data into respective training, validation and test sets. In order to ensure intraday prediction, which means that using trading days $t-p$, $t-(p-1)$, ..., $t-1$ we predict the price value of financial instrument on the t day, we can use the methodology of **rolling window** or commonly named as **sliding window**.

In this approach we define static time-series interval, which will serve as a hybrid train-validation time interval. Let's define the overall data interval being between T_1 and T_1+M breakpoints, meaning the overall length of the interval is M *(global interval)*. On this interval reside two frames of variables – possibly multivariate $Feature$ which serves as a input variable and univariate $Target$, which accumulates values of the predicting variable. Further, we define parameters $p < M$, which is the length of training subwindow *(look-back parameter)* and w, which denotes time units beyond which we aim to determine predictions. For the purposes of this thesis and its intraday prediction methodology we define $w = 1$, but in general auto-regressive case $w \geq 1$ is acceptable.

For the purposes of machine learning context of feature and target variables, we denote single pieces of input and output variables as $\mathbf{X}_i$ and Y_i respectively. Using the defined parameters, division of the global interval is performed as:

for $i = T_1$ to $T_1 + M - (p + w)$ **do**
$X_i \leftarrow Feature[i, i + p]$
$Y_i \leftarrow Target[i + p + 1, i + p + w]$
end for

From the rolling window algorithm pseudocode above a few imporant properties of this method could be derived. Firstly, respective intervals are overlapping and the shift between respective $\mathbf{X_i}$, $\mathbf{X_{i+1}}$ and its Y_i, Y_{i+1} is w, in our particular case the single time unit. Secondly, by ensuring overlapping, we maxime number of possible training, thus validation, data points making the training-validation process possibly more robust and general.

This methodology could be straightforwardly implemented to the stated problem. Further, it is sufficient to define process of maintaining chronology in the data.

2.3 Data chronology and prediction methodology

Approach, implemented in the thesis, uses prediction based on historical developments of selected feature variables from financial and macroeconomical markets. In order to implicitly simulate flow of time in timeseries we introduce particular backward shifts, using which we create setup of the modelling according to above defined constraints.

Rigorously speaking, assume that method capable of predicting closing price in the day t is a function, where input is movements of selected features in days $t - 1$, $t - 2$, ..., $t - p$. Let's denote the function f. Straightforwardly, the overall setup in general, but the most importantly in the thesis, could be defined as:

$$f(\mathbf{X}_{[t-p,t-1]}) = \hat{Y}_t, \tag{2.1}$$

where $\mathbf{X}_{[t-p,t-1]} \in \mathbb{R}^{p \times t}$, $p > 1$ is the look-back parameter and $t \geq 1$, number of feature variables used, denotes historical developments of selected features in the interval $t - p$ till $t - 1$ and $\hat{Y}_t$ denotes the targeting price prediction of the instrument in time t.

In accordance with the algorithm adapted for the rolling window data derivation, it is wise to implement data preparation in accordance with it. Aligned with the (2.1), when predicting particular instrument's price in time t, we require having features' developments through interval of $t - p$ till $t - 1$. Maintaining this, in terms of implementation, for every single unit of time in given global interval and for every single feature, we attach its historical development up to the look-back parameter p. In terms of algorithm:

$FEATURES \leftarrow [F_1, F_2, ..., F_t]$
for $i = 1$ to t **do**
  for $q = 1$ to p **do**
    $F_{i,q} \leftarrow F_i$ shifted by q
  end for
end for

where F_i, $i \in \{1, 2, ..., t\}$ denotes single features selected for the model and $F_{i,q}$ denotes shift of the i-th parameter backwards by q units. With such shifted data series for every single feature, we build data points available in single days of the global interval. In rigorous manner, every single trading day t has attached processed features respective to it as:

$$\forall i \in \{1, 2, ..., t\}\, \forall q \in \{1, 2, ..., p\} : \mathcal{T}_t \leftarrow F_{i,t-q},$$

where $\mathcal{T}_t$ denotes universe of model input data available at day t. By combining this algorithm and preparation of rolling window samples we obtain datasets in accordance with safe work with time series, in terms of chronological sequencing, and suitable for applying predictions based on (2.1).

2.4 Suitable features in modeling

The amount of variables, which are directly or indirectly being generated together with day-to-day market running, is quite enormous and one should consider subsetting them for the machine learning purposes wisely. In this subchapter we aim to show and justify process of our general feature selection, mostly based on apriori knowledge of processes within financial markets and variables' most likely effect on stock markets.

For the sake of completness, data were obtained from the API directly fetching data available on Yahoo Finance – available through Python library abstraction as `yfinance` library [1], [13].

2.4.1 Stock indices

In order to process current general state of the stock exchange we researched possibilities of implementing data regarding price development of certain index funds. **Index fund** is a abstract concept of accumulating information regarding single stock assets into one unified asset, mostly using predefined weighted average of prices of stocks included.

Fund is specifically designed to follow and replicate performance of given subset of stocks. Taking this into account, by including such well chosen funds, which copy

broader and various scope of stocks on given market, we may obtain valuable information regarding current state of the US market.

For these purposes we decided to include into our database two major stock indices. General index **Standard's and Poor 500**, commonly known as S&P 500, is an index including 500 biggest US-traded stocks based on total market capitalization. Index consists of wide scope of various segments making it robust and universal in terms of mirroring overall US market state.

Taking into account the scope of this thesis – Information technology segment, we decided to include index **NASDAQ Composite**, better known as NASDAQ. The main difference to S&P 500 is its segment focus. NASDAQ focuses mainly on tech and IT companies, making it possibly a valuable piece of data to our database of feature data.

2.4.2 Commodities

Commodities denote mainly raw resources and materials, which are traded mainly in uniform quality and large quantities. Its fundamental role in the market comes from the fact, that most of these assets play essential part in further processings, consumption and industry. Matsumoto et al. suggest, that commodity market plays a vital part in whole financial segment, which suggests, that including such assets in database might be sufficient [8].

For the purposes of this thesis we included data regarding two major commodities – **crude oil** and **gold**. Even though, there might not be a clear connection to the IT sector, we believe that such data would bring valuable information about the general state in the whole global economy.

Oil plays a vital role in *keeping the world in motion* and a solid majority of world's activities depend on its presence and availability. This might imply, that these dependecies on the global market would be projected into price movement of the commodity.

Last but not least, we introduce gold price movements into the database. With the wide range of reasons, gold still remains as one of the most prominent securities. Most modern currencies are dependent on system, where gold acts as a security stabilizing currency movements and limiting governments issuing money. On top of that, governments rely on physical gold as a reserve in times of crises [10]. Taking these reasons into account, by including gold in the database of features, we assume obtaining valuable information regarding world's economy state, crisis potential or investors' sentiment etc.

2.4.3 Currency pairs - FOREX

Currency pairs and subsequently **Foreign Exchange Market** (FOREX) denote specific market, where the main traded goods are currencies. Traders involved in the market speculate on certain exchange pairs volatility and bet on them in order to make profit. Market itself determines price of every possible currency pair, thus making acquisition of certain amount of currency possible through various ways – currency X could be traded through currencies Y, Z, etc., making the market far more speculative than others. By the volume of instruments traded, this decentralized market is by far the largest market in the world [12].

Being fully decentralized, meaning prices being determined on this market are accepted worldwide, these data could be considered as a sufficient indicator of currency relative strength and by that indicator of particular government's and economical competitivness of certain country. Taking this into account, we assume, that including such data inside our database would bring valuable information about international competitivness of US economy towards other major economies and quantitative information about relations among major economies, which is most likely important predictor for stock prices' movements.

By our focus on US market, we chose including currency pairs with American dollar (**USD**) in all pairs. We chose to include so called **major currency pairs**, with respect to their overall traded volume on daily basis, according to reports published by **Bank for International Settlements** and to market's conventions [3], [12].

In particular, included pairs, together with USD, are Euro **(EUR/USD)**, Japanese yen **(JPY/USD)**, British pound sterling **(GBP/USD)**, Swiss frank **(USD/CHF)**, Australian dollar **(AUD/USD)**, Canadian dollar **(USD/CAD)** and New Zealand dollar **(NZD/USD)**.

2.4.4 Other market indicators

Apart from the introduced indicators and general assets price movement data above, there is still a vast group of possibly valuable features, which do not fit into above defined features' groups.

Chapter 3

Predictive analyses on IT segment

This chapter will be dedicated to certain results obtained during training and validation of several algorithms and approaches on the data. We plan to analyze results brought by included algorithms in first chapter in detailed manner.

Chapter 4

Intraday optimization of portfolio based on model's prediction

This chapter will deal with the topic of using the obtained data predictions *(preferably from the most precise model)* to optimize weights of single stock assets in hypothetical investment portfolio. For these purposes, we introduce Markowitz portfolio theory and the optimization problem coming out of it. With such foundations we analyze results in terms of profit and risk by our models.

Conclusion

This section will be dedicated to concluding our results, accumulating them into a concise form, evaluating successfulness of completing given goals and suggesting future tweaks in order to make thesis' idea more robust and practice-friendly.

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Appendix A: Source code of the model

Appendix attached to this thesis contains all the source codes used in analyses introduces in the thesis. All the source codes and results are freely available as a Git repository at `https://github.com/sugy-w/Bachelor-Thesis`.

Please note, that all the contents of this repository is being shared only for academic purposes and its modifying, redistribution or commercial use is strictly prohibited by the included licenses.